

OSNIAS CLEARING

Funding & Technical Roadmap — up to the security audit

Version 0.5 | October 2026 | Distribution version

OBJECTIVE

Structure the next phase of Osnias Clearing into fundable technical building blocks, building on the foundation already in place — Books I–VII documentation, testnet deployments, regulatory work and ecosystem contacts — up to the integrated demonstrator, security hardening and independent audit.

Overview

Completed	Draft testnet — to rework	To develop	To audit
B0 Reference architecture Technical corpus Books I–VII Clearing rules Regulatory pre-assessment Exploratory ecosystem contacts	B1 Clearing engine B2 Sei ledger B3 EVM escrow & controller B4 Osnias-ID / Node Registrar B5 1W time oracle B7 Node registry B10 Clearing platform (single node)	B5 4W / 13W / solvency oracles B6 ISO 20022 B7 Pilot O-POP B8 Reconciliation B9 Security hardening B10 Multi-node demonstrator	B11 Audit readiness B12 Independent audit Remediation / Re-test

KEY COMPLETED: foundation available | **DRAFT TESTNET:** working on Sei / Sepolia, to be reworked into clean code (Cursor AI → Foundry → Slither) | **TO DEVELOP:** specified or planned block | **TO AUDIT:** preparation, audit, remediation.

Funding principle

The aim is not to fund Osnias Clearing as a single monolithic block, but to start from the existing foundation and identify, for each work package, the maturity level, deliverables, budget, dependencies and the most relevant national or European funding scheme.

Founder, market and regulatory framework

Project founder

Osnias Clearing was conceived, designed, organised and documented by its founder, Denis Bouzon, with 100 % self-funding to date. The project has now reached the stage where it needs support to complement the founder's skills for the next steps: clean code rework, security, the multi-node demonstrator and audit.

Market: a solution that does not yet exist

Osnias offers companies a netting solution that does not exist today. Multilateral netting is a mechanism proven for decades, but reserved for interbank and market infrastructures: as an order of magnitude, CLS and Clearstream together process around EUR 4,300 billion of notional volume per day, i.e. more than EUR 1 million billion per year. Recurring commercial obligations between companies have no equivalent. The aim is therefore not to take a share of an existing market, but to open this mechanism to inter-company flows.

Regulatory framework and risks

The main risk is regulatory: qualification of the infrastructure and of the clearing token. Pre-assessment is under discussion with the AMF; the files submitted to the CSSF (Luxembourg) and the AFA (Andorra) are awaiting discussion. The sector is waiting for this pre-assessment before any commitment: until then, exchanges with potential node operators remain exploratory.

The following design principles limit risk by construction (details: osnias-clearing.com/faq/):

Principle	Scope
Non-custodial	Osnias holds neither user funds nor private keys or seed phrases; collateral is escrowed separately on Ethereum.
No credit, overdraft or leverage	Execution invariant: instantaneous net exposure ≤ available collateral; any obligation exceeding it is rejected by the code.

Principle	Scope
Individualised collateral	No common guarantee fund: risk stays with each participant.
No central novation	Osnias is not designed as a CCP; distributed multilateral netting, final qualification to be determined with the authorities.
Node responsibilities	Admission, blocking and exit of participants are decided by the node manager, under its local framework.
No PFMI claim	No PFMI certification or compliance is claimed; the architecture refers to it for risk management.

Founder's estimate: around EUR 200k over 6 months for packages 1 to 4 (details: "Indicative timeline and budget").

Foundation already in place before the funding phase

The project is not starting from scratch. The work already completed forms the foundation from which technical work packages B0 to B12 can be qualified, budgeted and matched to suitable funding schemes.

Area	Elements already in place	Status / scope
Technical documentation	Structured Books I–VII technical corpus covering the architecture, Osnias-ID, clearing cycles, security and rules, onboarding and the O-POP / RPC architecture. Related position papers and public documentation.	Documentation baseline available; to be kept aligned with the technical baseline.
Draft testnet development	Eight working draft components: 1W time oracle, Sei registry (identity), Sei controller and ORUSD ledger on Sei Atlantic-2 (Sei contracts verified); Node Registrar, EVM registry, EVM escrow and controller on Ethereum Sepolia; ORUSD Clearing Desk platform online. Details: osnias-clearing.com/deployment/ and /end-user/ .	Draft development on testnet; code to be reworked (Cursor AI → Foundry → Slither) before audit.
Regulatory pre-assessment	France — AMF: memorandum and regulatory pre-assessment file. Andorra — AFA: bilingual pre-assessment file submitted. Luxembourg — CSSF: bilingual pre-assessment file submitted.	Preparatory regulatory work already completed across three jurisdictions.
Ecosystem & third parties	Exploratory contacts with banking, fintech and infrastructure / technology players, as well as other institutional stakeholders.	Exploratory contacts and exchanges only; no partnership or consortium is presented as committed.
Outreach & traceability	Public website, summary FAQ, FR / EN documentation, architecture diagrams and technical publications enabling a third party to assess the project quickly before accessing detailed documents.	Qualification material for funders, technical partners and authorities.

For a funder: the protocol is in draft development on testnet, with the main components deployed. The phase to be funded covers clean rework of the code, integration, the multi-node demonstrator, security hardening and the independent audit.

Components already developed in draft (testnet)

The components below are deployed on testnet in draft development: they will be reworked through a clean development chain before the baseline is frozen and audited.

Component	Network	Status	What works
1W time oracle	Sei Atlantic-2	Deployed, verified	Weekly cycle, MINT / BURN / CLEARING / P2P windows, cycle number and start block.
Sei registry (identity)	Sei Atlantic-2	Deployed, verified	Identity of participants and nodes on the clearing side.
Sei controller	Sei Atlantic-2	Deployed, verified	Links oracle, registry, clearing token and the Sepolia EVM chain.
ORUSD clearing ledger	Sei Atlantic-2	Deployed	USD ledger positions; MINT, BURN, P2P and transfer instructions.

Component	Network	Status	What works
Node Registrar	Ethereum Sepolia	Deployed	Registration of KK-NNN nodes (first node: 90-017).
EVM registry	Ethereum Sepolia	Deployed	KK zone registry on the EVM side.
EVM escrow & controller	Ethereum Sepolia	Deployed	Collateral escrow and control on the EVM side.
Clearing platform (Clearing Desk)	Sei Atlantic-2	Online, draft	Wallet connection, ledger position, oracle state, function availability by window, registry.

Addresses, creation transactions and verification parameters: osnias-clearing.com/deployment/ · platform: osnias-clearing.com/end-user/

Clean rework chain for the draft code

The tool matters less than the framing: each component is reworked in strict conformity with the technical specifications of Books I to VII.

- Cursor AI — assisted rewrite of the draft code from frozen specifications, consistent conventions, NatSpec documentation.
- Foundry — unit tests, fuzzing, invariant tests and multi-participant simulations.
- Slither — static analysis, detector review and remediation, dependency review.
- Verified testnet redeployment, then freeze of the baseline handed over to security hardening and audit.

Rework chain: from draft testnet to an auditable baseline



Figure 1 — Rework chain: from draft testnet to an auditable baseline

Documentation corpus — Books I to VII

The technical corpus is published as seven books, each available in French and English on the public website and archived on Zenodo with a permanent DOI.

Book	Title	Release	Public page	Zenodo DOI
I	Osnias-ID	3.0	osnias-clearing.com/osniasid/	10.5281/zenodo.22978105
II	Architecture	2.0	osnias-clearing.com/architecture/	10.5281/zenodo.23012840
III	Clearing rules	1.1	osnias-clearing.com/rules/	10.5281/zenodo.22922199
IV	Clearing cycles 1W / 4W / 13W	—	osnias-clearing.com/cycle/	10.5281/zenodo.22979580
V	Security topology & resilience	2.0	osnias-clearing.com/security/	10.5281/zenodo.22981575
VI	Onboarding via a node	1.0	osnias-clearing.com/onboarding/	10.5281/zenodo.23010805
VII	O-POP Osnias — mainnet architecture	1.0	osnias-clearing.com/pop/	10.5281/zenodo.23055994

Each public page offers the FR / EN versions and the book PDFs. Releases as at the distribution date of this document.

B0 — Reference architecture

STATUS **COMPLETED**

Available elements:

- Separation of clearing / settlement / custody functions.
- Architecture of the canonical clearing ledger and its interface with the EVM layer.
- Netting cycles, operating rules and main invariants.
- Existing documentation corpus: Books I–VII technical documentation, position papers, operating rules, functional documentation and regulatory work.

- Architecture doctrine: Osnias core built on standard EVM primitives, network dependencies confined to adapters; Sei remains the current reference deployment, and local O-POPs remain standardised and replicable.

Work to be carried out / consolidated:

- Freeze a versioned technical baseline for integrated development and the future audit scope.

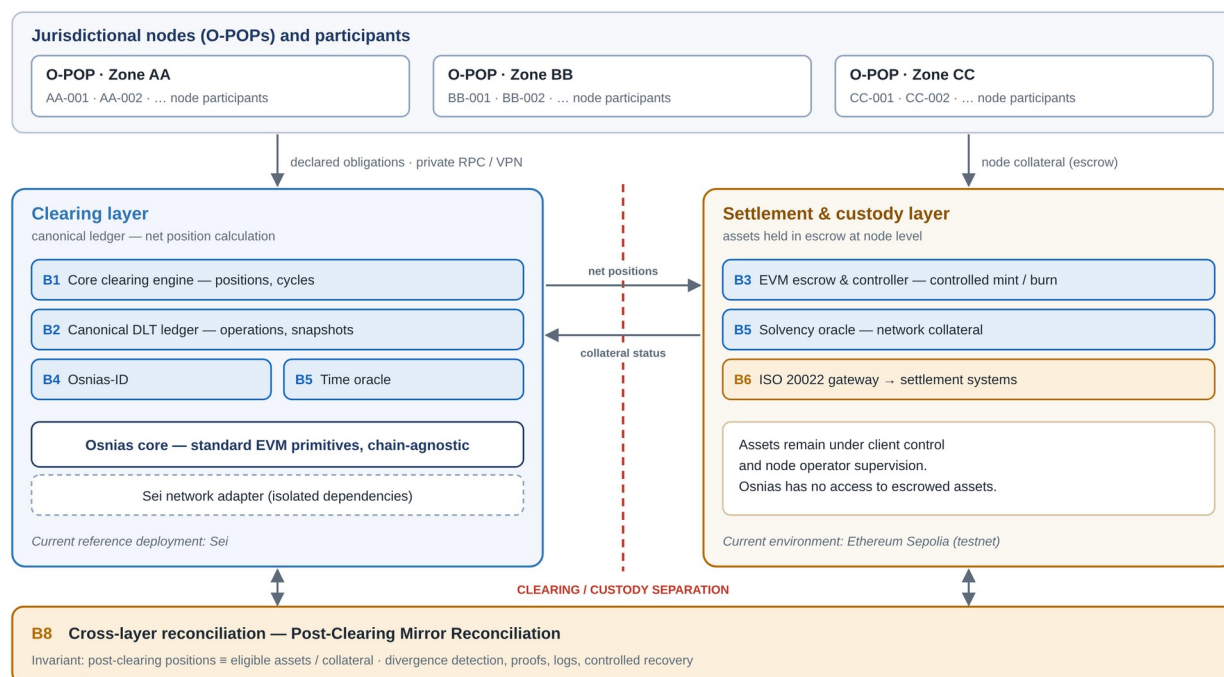


Figure 2 — Target architecture: clearing layer, settlement & custody layer and cross-layer reconciliation

B1 — Core Clearing Engine

STATUS **DRAFT TESTNET — TO REWORK**

Available elements:

- On-chain logic working in draft on Sei: controller, ORUSD ledger, MINT / BURN / CLEARING / P2P windows.
- MINT / BURN / CLEAR states, snapshots at each cycle transition and freeze rules.
- Global position-consistency invariant specified.

Work to be carried out / consolidated:

- Rework the draft code through the Cursor AI → Foundry → Slither chain.
- Assemble an integrated, reproducible reference multilateral netting engine.
- Automate unit tests, fuzzing and multi-participant simulations; document edge cases.

B2 — Canonical DLT ledger — Sei reference deployment

STATUS **DRAFT TESTNET — DEPLOYED AND VERIFIED**

Available elements:

- Sei registry (identity), Sei controller and ORUSD ledger deployed on Sei Atlantic-2; Sei contracts verified (exact match).
- Time oracle and position consolidation.
- Ledger of clearing operations, viewable through the clearing platform.

Work to be carried out / consolidated:

- Clean rework of the code (Cursor AI → Foundry → Slither) and verified redeployment.
- Test recovery, replayability, failure, load and write consistency.
- Formalise interfaces and events; isolate Sei dependencies in a dedicated adapter so that future EVM migration or multi-deployment remains possible without rewriting the core.

B3 — EVM Settlement & Collateral

STATUS **DRAFT TESTNET — DEPLOYED**

Available elements:

- EVM escrow and controller deployed on Ethereum Sepolia.

- EVM zone registry; solvency checks and controlled mint / burn.

Work to be carried out / consolidated:

- Clean rework (Cursor AI → Foundry → Slither) and contract verification on the explorer.
- Integrate the EVM layer with the canonical ledger; strengthen fuzzing and solvency invariants.
- Test adversarial behaviour, errors and recovery procedures.

B4 — Osnias-ID

STATUS **DRAFT TESTNET — DEPLOYED**

Available elements:

- Osnias-ID format (network / node, checksum, anti-collision) — Book I, Release 3.0.
- Identity layer on Sei; Node Registrar and EVM registry on Sepolia; first node registered (90-017).
- Fail-closed behaviour specified.

Work to be carried out / consolidated:

- Clean rework of the code; integrate Osnias-ID across all components and messages.
- Test collisions, checksum, multi-network and fail-closed scenarios.

B5 — Time & solvency oracles

STATUS **PARTIAL — 1W ORACLE IN DRAFT**

Available elements:

- 1W time oracle deployed and verified on Sei, read live by the clearing platform (cycle, window, start block).
- Solvency oracle specified for network collateral control.

Work to be carried out / consolidated:

- Develop the 4W and 13W oracles and the solvency oracle.
- Define redundancy, thresholds and failure policy; test unavailability, delay, outlier values, freeze and recovery.

B6 — ISO 20022 interoperability

STATUS **SPECIFIED — TO DEVELOP**

Available elements:

- Functional architecture and interface requirements with nodes and settlement systems identified.

Work to be carried out / consolidated:

- Define the mapping of Osnias events ↔ ISO 20022 messages.
- Develop orders, confirmations, statuses and error handling.
- Build end-to-end simulated settlement scenarios.

B7 — Jurisdictional O-POPs & Nodes

STATUS **PARTIAL — NODE REGISTRY IN DRAFT**

Available elements:

- O-POP topology by KK zone and KK-NNN nodes; on-chain node registration working in draft (Node Registrar).
- Private RPC, VPN tunnels, logging and monitoring principles.
- Replicable jurisdictional O-POPs, connected to the same Osnias core and operable according to local requirements.

Work to be carried out / consolidated:

- Deploy a first pilot O-POP and produce a reproducible package: configuration, network plan, hardening, monitoring, documentation and acceptance testing.
- Test network security, connection loss, latency, recovery and service continuity.

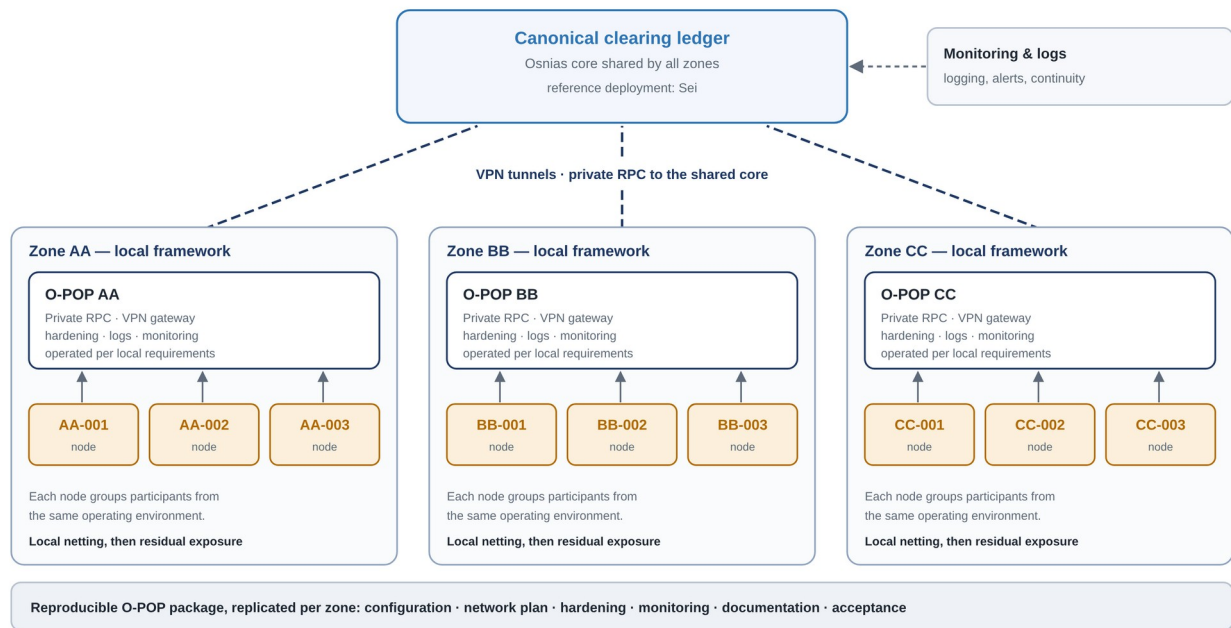


Figure 3 — Jurisdictional O-POP and node topology (illustrative zones and codes)

B8 — Cross-layer reconciliation

STATUS **SPECIFIED — TO DEVELOP**

Available elements:

- Post-Clearing Mirror Reconciliation principle documented.
- Consistency invariant between post-clearing positions and eligible assets / collateral.

Work to be carried out / consolidated:

- Automate divergence detection, proofs, logs and corrective procedures.
- Test desynchronisation scenarios and controlled recovery.

B9 — Security Hardening

STATUS **TO BE FUNDED**

Work to be carried out / consolidated:

- Static analysis and dependency review.
- Foundry, fuzzing and invariant testing.
- Access controls, adversarial scenarios, economic attacks and DoS.
- Review of the threat model and operating procedures.
- Exit criterion: no known critical vulnerability before handover to the auditor.

B10 — Integrated Testnet Demonstrator

STATUS **PARTIAL — SINGLE-NODE PLATFORM IN DRAFT**

Available elements:

- ORUSD Clearing Desk platform online on Sei testnet: ledger position, oracle state, MINT / BURN / P2P / instruction windows and functions, registry.

Work to be carried out / consolidated:

- Extend to a multi-node demonstrator: simulated participants, several nodes, multilateral clearing, collateral, settlement, Osniash-ID, oracles and reconciliation.
- Run several complete cycles, nominal and failure scenarios.
- Produce evidence, logs, metrics and reproducible documentation.
- Measure the netting rate (share of gross settlements eliminated) on simulated flow datasets.

B11 — Audit Readiness

STATUS **TO BE FUNDED**

Work to be carried out / consolidated:

- Code freeze and reference version.
- Final architecture, specifications, threat model and list of invariants.

- Test dossier, coverage, contract documentation and operating procedures.
- Precise definition of the scope handed to the auditor.

B12 — Independent security audit

STATUS **TO BE FUNDED**

Work to be carried out / consolidated:

- Independent audit of smart contracts and critical components.
- Review of invariants and attack surfaces.
- Audit report, remediation plan, fixes and re-test.

Proposed grouping into fundable work packages

Package	Scope	Blocks	Main deliverable
PACKAGE 1	Core Protocol	B0 to B4	Clean rework of the draft (Cursor AI → Foundry → Slither): clearing engine, chain-agnostic EVM core, Sei adapter, EVM settlement and Osnias-ID.
PACKAGE 2	Infrastructure & Interoperability	B5 to B7	Oracles, ISO 20022, jurisdictional O-POPs and nodes.
PACKAGE 3	Integrated Demonstrator	B8 to B10	Reconciliation, security hardening and multi-node demonstrator.
PACKAGE 4	Security Audit	B11 to B12	Audit readiness, independent audit, remediation and re-test.

Overall dependency: Package 1 → Package 2 → Package 3 → Package 4. Some work can be carried out in parallel once the relevant interfaces and invariants have been stabilised.

Proposed exit criteria per work package

Package	Exit criteria (proposed, to be validated)
PACKAGE 1	Code reworked and compliant with the specifications; Foundry tests (unit, fuzzing, invariants) passing; no unresolved high-severity Slither detector; contracts redeployed and verified.
PACKAGE 2	1W / 4W / 13W and solvency oracles operational; ISO 20022 messages sent and acknowledged; one pilot O-POP accepted.
PACKAGE 3	Several complete cycles on at least three nodes; netting rate measured on simulated flows; no known critical vulnerability.
PACKAGE 4	Independent audit report; remediation completed and re-tested; baseline frozen for the pilot.

Deployment sequence

1. Testnet (M1 – M4): clean rework of the draft code (Cursor AI → Foundry → Slither), integration of the building blocks, multi-node demonstrator.
2. Security audit (M5 – M6): independent audit, remediation and re-test; audited baseline frozen.
3. Regulatory approval: file submitted with the audited baseline to the competent authority (AMF; CSSF and AFA depending on the territory), then validation, authorisation or non-objection. Duration depends on the regulator.
4. Mainnet: deployment only after regulatory approval.
5. Operation.

The funded work remains useful whatever the final regulatory qualification: the chain-agnostic core, the clearing / custody separation and the audited baseline can be reused under any status retained.

Indicative timeline and budget

Founder's estimate: around EUR 200k over 6 months. Proposed breakdown per package, to be refined with detailed costing.

Package	Period	Estimated budget	Type of expenditure
PACKAGE 1 — Core Protocol	M1 – M2	EUR 55k	Clean rework of the draft code (Cursor AI → Foundry → Slither), testing, verified redeployment.

Package	Period	Estimated budget	Type of expenditure
PACKAGE 2 — Infrastructure & Interoperability	M2 – M3	EUR 50k	4W / 13W and solvency oracles, ISO 20022 gateway, pilot O-POP (servers, VPN, monitoring).
PACKAGE 3 — Integrated Demonstrator	M3 – M4	EUR 45k	Reconciliation, security hardening, 3-node demonstrator, netting-rate measurement.
PACKAGE 4 — Security Audit	M5 – M6	EUR 40k	Independent contract audit, remediation, re-test.
Coordination, regulatory advice, contingency	M1 – M6	EUR 10k	Coordination, preparation of the regulatory file, contingency (5 %).
Total	6 months	EUR 200k	

Months counted from the start of funding (M1). The budget covers testnet, audit and preparation of the regulatory file; regulatory approval, mainnet and operation follow, outside these 6 months.

Sequence: testnet and audit over 6 months, then regulatory approval, mainnet and operation

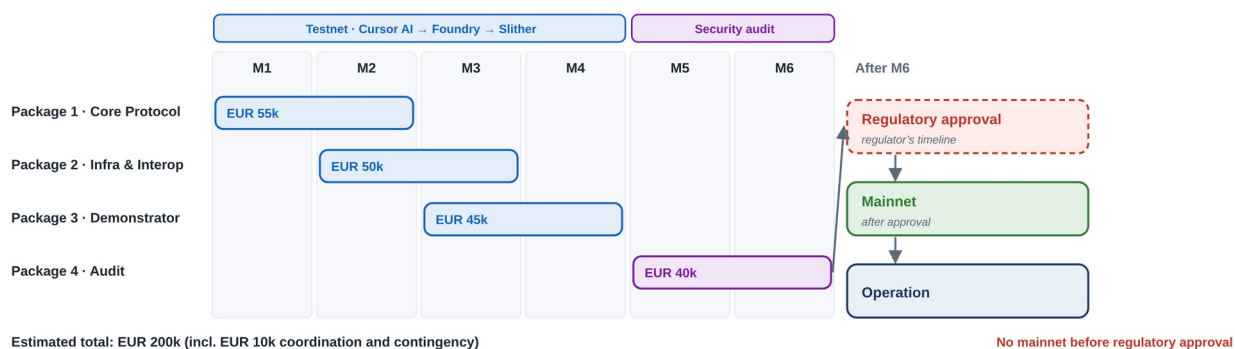


Figure 5 — Testnet and audit over 6 months, then regulatory approval, mainnet and operation

Pathway: from the existing foundation to the institutional pilot

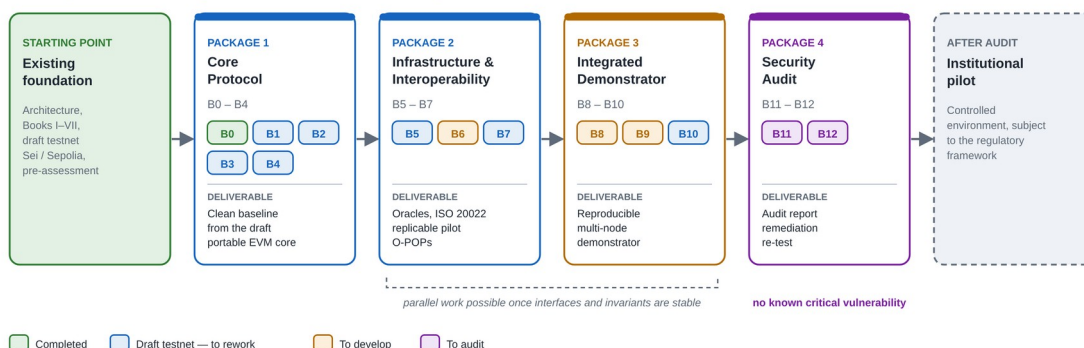


Figure 4 — Work-package pathway, from the existing foundation to the institutional pilot

Expected work with an innovation funding partner

- Qualify the technology readiness level of each building block, without imposing a single TRL on the whole project.
- Assess the budget and eligible expenditure for each work package.
- Identify suitable national or European schemes for each stage.
- Take the existing assets — documentation, testnets, regulatory work and ecosystem contacts — into account when assessing the maturity and risk of each package.
- Identify any partners required for collaborative calls.
- Build a consistent timeline across submissions, development, demonstrator and audit.

Purpose: build a progressive, verifiable funding and industrialisation pathway in which each work package delivers a demonstrable technical output and prepares the next, up to the independent security audit and then the institutional pilot.

OSNIAS CLEARING

Funding & Demonstrator — Executive One-Pager

Executive summary | October 2026 | Funding qualification to be opened package by package

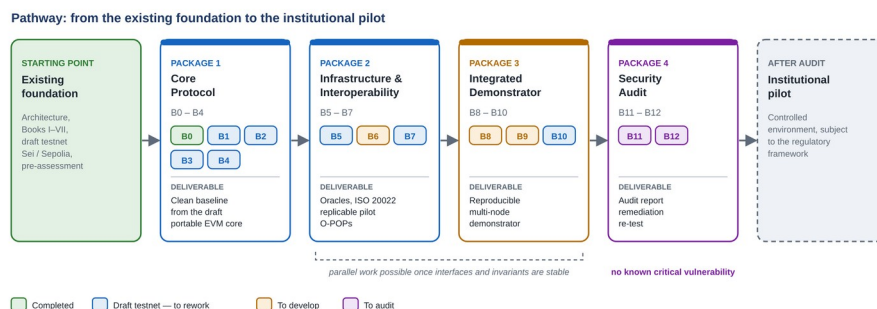
STARTING POINT

The project is not starting from zero: documented architecture, a seven-book corpus, a protocol working in draft on testnet (clearing platform online on Sei), regulatory work and initial ecosystem exchanges. The next step is to fund clean rework of the code, integration, the multi-node demonstrator and audit preparation.

1. Assets already in place

- Reference architecture: separation of clearing / settlement / custody, netting cycles, invariants and technical governance.
- Portability doctrine: Osnias core in standard EVM, network dependencies isolated in adapters; Sei remains the current reference deployment with no structural lock-in of the core.
- Books I–VII technical documentation corpus, FR/EN public documentation, diagrams, FAQ and protocol rules.
- Working draft on Sei Testnet and Ethereum Sepolia: 1W oracle, registries, controllers, escrow, Node Registrar and an online clearing platform — to be reworked cleanly (Cursor AI → Foundry → Slither) before audit.
- Regulatory pre-assessment work: bilingual files submitted to the AFA (Andorra) and the CSSF (Luxembourg), work under way with the AMF (France).
- Project conceived, documented and 100 % self-funded by its founder; it opens multilateral netting, today reserved for interbank and market infrastructures, to companies.

2. Four fundable work packages up to the audit



3. Scope of work with the funding partner

- Qualify the maturity of each building block and its entry / exit TRL, without imposing a single TRL on the project.
- Identify the French or European scheme suited to each package, as well as genuinely eligible expenditure.
- Build the budget, the submission timeline and, where needed, the appropriate consortium or technical partners.
- Sequence funding so that each package delivers a demonstrable output and prepares the next, up to the security audit.

4. Parameters to be qualified

Budget and timeline: founder's estimate (see "Indicative timeline and budget"). TRL and schemes: to be qualified at the first funding discussion.

Package	Entry / exit TRL	Budget	Scheme	Timeline
1	To be qualified	EUR 55k	To be qualified	M1 – M2
2	To be qualified	EUR 50k	To be qualified	M2 – M3
3	To be qualified	EUR 45k	To be qualified	M3 – M4
4	To be qualified	EUR 40k	To be qualified	M5 – M6

OBJECTIVE

Move from the existing technical and regulatory foundation to an integrated, multi-node, documented demonstrator ready for independent audit, through a progressive funding and industrialisation pathway rather than monolithic funding.

Post-audit outlook: after the independent audit and remediation, the next phase will target an institutional pilot in a controlled environment, subject to the regulatory framework and the partners selected.

Distribution version — indicative budget and timeline; TRL and schemes to be qualified.